

0 4 . 1 Draw the **general** structure of an amino acid.

[1 mark]

0 4 . 2 Which statement describing protein structure is correct?

Tick (✓) **one** box.

[1 mark]

Proteins with a quaternary structure have one or more polypeptide chains.

☐

Disulfide bridges are present in all proteins with a tertiary structure.

☐

The secondary structure of a protein is maintained by ionic and hydrogen bonds.

☐

The primary structure of a protein is the sequence of amino acids joined by peptide bonds.

☐

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Antibodies are proteins produced and secreted by plasma cells.

Once activated, a single B cell divides and produces 4000 plasma cells.

Each one of these plasma cells produces 2000 molecules of antibody Q per second.

A human male has a blood volume of 5.4 dm^3

Calculate the total number of molecules of antibody Q per cm^3 of blood 24 hours after the production of these plasma cells.

[2 marks]

Answer = _____ molecules of antibody Q per cm^3 of blood

Question 4 continues on the next page

0 4 . 4

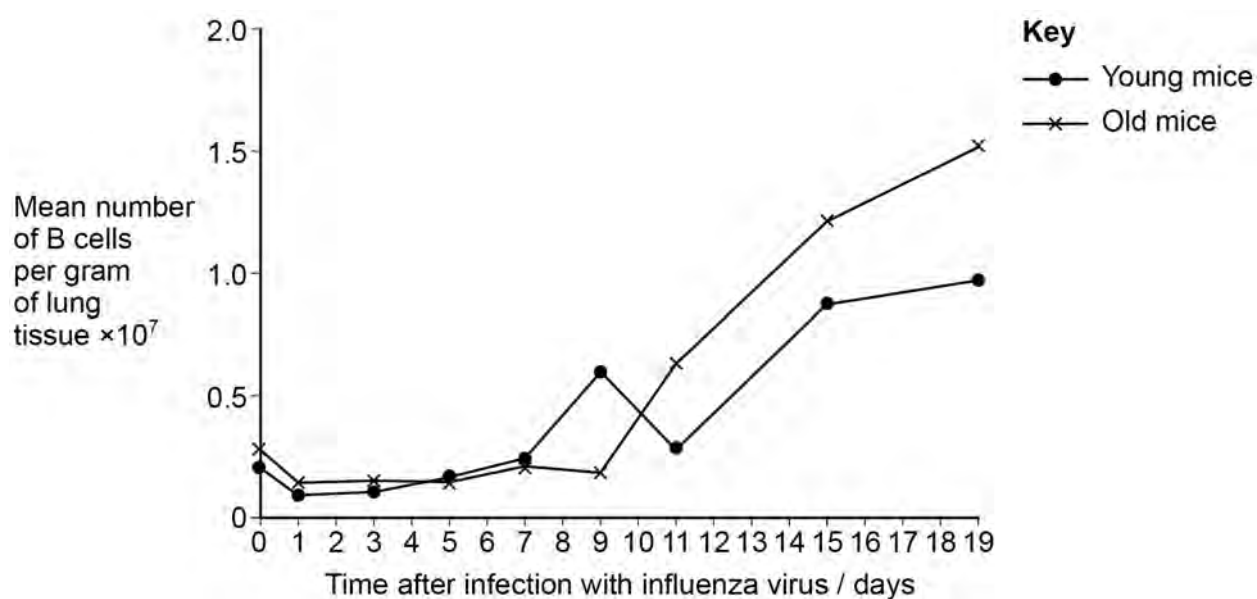
Scientists investigated the effect of age on the immune response in healthy female mice.

The scientists used the following method.

1. Infected a large number of young and old mice with the same strain of influenza virus.
2. Removed the lungs from 6 young and 6 old mice on the day of infection (day 0).
3. Repeated step 2 on 8 different days after infection.
4. Determined the mean number of B cells present per gram of lung tissue for young and old mice.

Figure 3 shows the scientists' results.

Figure 3



The scientists decided that further investigations would be required to conclusively determine the effect of age on the immune response in healthy female mice.

Using all the information provided, suggest **three** reasons for the scientists' decision.

Do **not** refer to statistical tests, repeats or sample size in your answer.

[3 marks]

1 _____

2 _____

3 _____

Turn over for the next question

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